

CEIT 436 – Project Refinement Plan

Digital Detective Academy

1. Brief Project Overview

Group Number: 12

Group Name: Digital Detective Academy Team

Group Members:

- Volkan Ulusoy – Technical Development, UI/UX Implementation
- Paloosha Sheikh – Instructional Design & Content Development
- Hina Hotak – Project Manager & Coordination

Project Description

Digital Detective Academy is a web-based educational platform designed to improve digital literacy and cybersecurity awareness among middle school students. The platform uses a story-driven learning approach where students interact with real-life digital scenarios and make decisions related to online safety.

The system currently includes multiple instructional modules, interactive questions, and scenario-based activities that help students understand cybersecurity concepts such as phishing, password safety, and responsible digital behavior.

Main Goal of the Project

The primary goal of the project is to create an engaging and accessible learning environment that helps middle school students develop essential digital literacy and cybersecurity awareness skills through interactive and story-based learning experiences.

2. Current Status & Limitations

Current Status

At the end of CEIT 435 course , the project successfully reached a fully functional prototype stage. The platform includes:

- A working web-based interface
- Story-based cybersecurity learning modules
- Interactive activities and decision-based questions
- A responsive and minimal design
- Basic usability testing conducted with middle school students

User testing results showed that students were able to navigate the platform independently and were highly engaged with the interactive activities.

What is Working Well

Several aspects of the platform performed well during testing:

- Interactive activities successfully captured students' attention.
- The story-driven structure made cybersecurity topics easier to understand.
- The navigation and layout were simple enough for students to use without guidance.
- Students were able to identify unsafe digital behaviors after completing activities.

Identified Limitations

Despite the successful prototype, several areas require improvement:

- Some CSS conflicts caused inconsistencies in visual elements.
- Certain game activities require functional improvements.
- Some content sections are longer than necessary for the target age group.
- The menu navigation could be improved to make the platform more intuitive.
- Visual consistency across pages can still be enhanced.

These limitations provide the basis for the refinement work planned for CEIT 436.

3. Refinement Goals for This Semester

The goal of this semester is to improve the quality, usability, and stability of the platform while refining the learning experience.

The following refinements are prioritized:

1. Technical Improvements

- Resolve CSS conflicts to ensure consistent styling across the platform.
- Improve the functionality of existing interactive games.
- Fix minor usability issues identified during user testing.

2. Design Improvements

- Redesign the navigation menu to make it more user-friendly.
- Improve layout consistency across pages.
- Enhance visual clarity and readability for younger learners.

3. Content Refinement

- Simplify longer text sections to better match the reading level of middle school students.
- Improve instructions within activities and games.
- Ensure better alignment between scenarios and learning objectives.

These refinements will help transform the prototype into a more polished and stable educational platform.

4. Planned Improvements & Features

The project team plans to implement the following improvements during CEIT 436.

1. CSS Conflict Resolution

Some interface elements currently show minor styling inconsistencies due to CSS conflicts.

Improvement:

- Review and reorganize CSS structure
- Standardize styles across pages

Rationale:

This will improve visual consistency and make the interface more professional and easier to maintain.

2. Game Content and Functionality Improvement

The existing educational games will be improved to ensure smoother interaction and clearer learning outcomes.

Improvement:

- Refine game logic and interactions
- Improve feedback messages after user choices
- Adjust game difficulty to match the target age group

Rationale:

Improving the games will enhance engagement and strengthen the learning experience.

3. Menu and Navigation Redesign

The current navigation structure will be simplified and redesigned.

Improvement:

- Create a more intuitive menu structure
- Improve button placement and navigation flow
- Make the interface easier to use for younger learners

Rationale:

A clearer navigation system will reduce confusion and improve usability.

4. Content Simplification

Some instructional sections contain longer explanations that may slow down younger learners.

Improvement:

- Shorten and simplify text sections
- Add clearer instructions and examples

Rationale:

Simplified content improves comprehension and keeps students engaged.

5. User Testing Plan (Brief Overview)

To evaluate the improvements made in the project, additional user testing will be conducted.

Target Users

- Middle school students
- Age range: 11–13 years

Testing Focus

The testing sessions will focus on:

- Usability of the improved navigation system
- Clarity of simplified content
- Functionality and engagement of updated games
- Overall user experience of the platform

Students will interact with the platform and provide feedback regarding their experience.

6. Timeline & Task Distribution

Timeline (High-Level Plan)

Weeks 1–3

- Review prototype and identify technical issues
- Plan design and content improvements

Weeks 4–7

- Resolve CSS conflicts
- Improve game functionality
- Simplify instructional content

Weeks 8–10

- Implement new menu design

- Improve interface consistency

Weeks 11–12

- Conduct user testing sessions

Weeks 13–14

- Analyze feedback and finalize refinements

Task Distribution

Volkan Ulusoy

- Technical development
- CSS conflict resolution
- Implementation of game improvements
- Interface adjustments

Paloosha Sheikh

- Instructional design
- Content simplification
- Learning activity improvements

Hina Hotak

- Project management and coordination
- Timeline tracking and documentation
- User testing organization and reporting

7. Project Tracking Link (CEIT 436 Update)

The project tracking document created during CEIT 435 will continue to be used during CEIT 436.

The link is:

<https://ahmtoz.github.io/Easeo/>